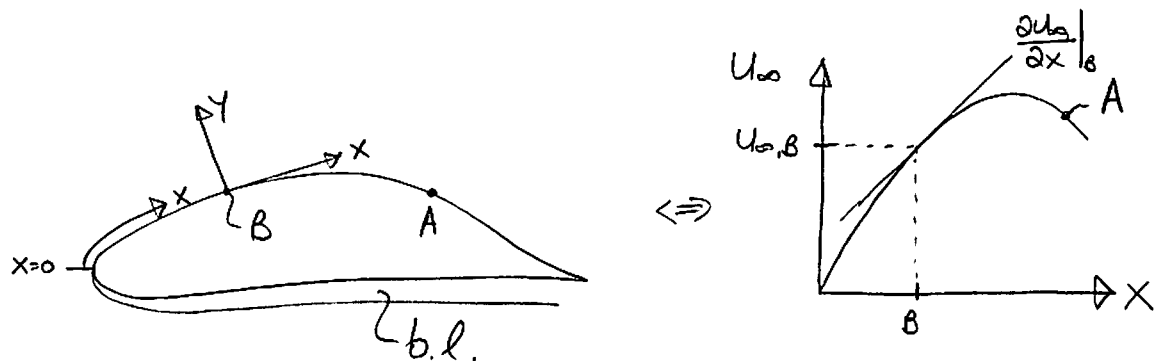


Local Similarity

Suppose we had an arbitrary shape with flow around it:



Assumption, conditions upstream have little influence on the b.l. behaviour at location x .

Thus the b.l. thickness is mainly a manifestation of local conditions.

This is only an approximate analysis. From our U_∞ profile, m can be calculated for a local wedge flow:

$$\left. \begin{aligned} U_\infty &= Cx^m \\ \frac{\partial U_\infty}{\partial x} &= Cm x^{m-1} = m \frac{U_\infty}{x} \end{aligned} \right\} m = \frac{x \frac{\partial U_\infty}{\partial x}}{U_\infty}$$

We assume the b.l. thickness at x is identical to that of a wedge flow with:

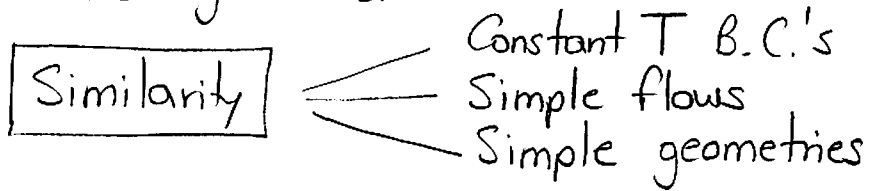
$$\beta = \frac{2m}{m+1}$$

We can use this to solve for wall shear stress: $\tau = \mu \frac{\partial u}{\partial y} \Big|_x$

We can also solve for when separation will occur $\tau \approx \mu \frac{U_\infty}{\delta(x)}$
($\beta = -0.1988$).

Integral Methods

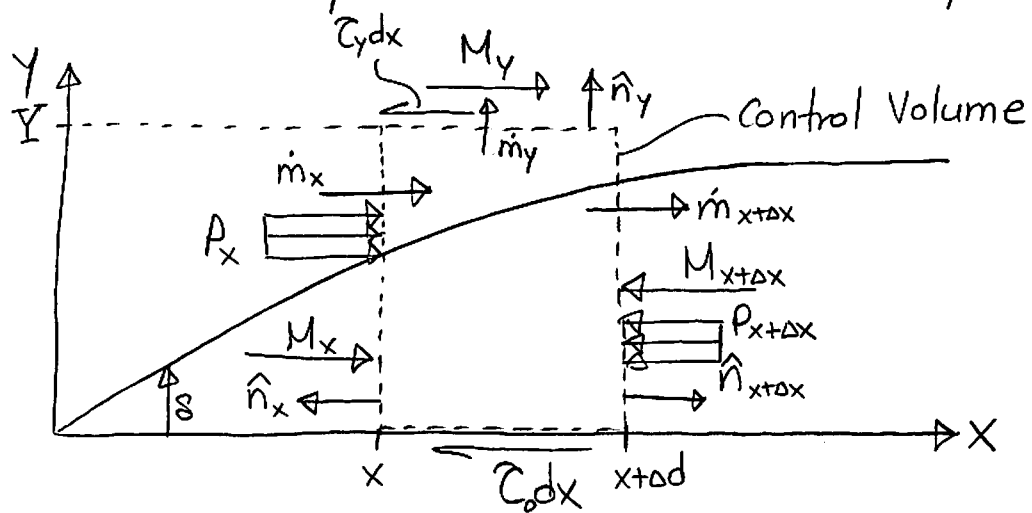
So far, we have only looked at similarity solutions, which are good for:



But what if $T = f(x)$? or what about $q''|_{y=0} = \text{constant}$?

For more complex flows and geometries, it turns out integral techniques are quite useful.

Let's look at how they work. Consider a boundary layer:



Looking at each of our terms for an x-momentum balance:

$$M_x = (\dot{m}u)|_x \Rightarrow \dot{m}|_x = \int_0^Y \rho u dy$$

$$M_x = \int_0^Y \rho u^2 dy|_x \quad (1)$$

$$M_{x+\Delta x} = (\dot{m}u)|_{x+\Delta x} = \int_0^Y \rho u^2 dy|_{x+\Delta x} \quad (2)$$

$$M_Y = (\rho v u)|_Y \cdot \Delta x \quad (3)$$

Now for our forces:

↙ Taylor Series Expansion

$$(\rho_x - \rho_{x+\Delta x})Y = \left[\cancel{\rho_x} - \left(\cancel{\rho_x} + \frac{\partial \rho}{\partial x} \Big|_x \Delta x + \frac{1}{2} \frac{\partial^2 \rho}{\partial x^2} \Delta x^2 + \dots \right) \right] Y \quad (4)$$

$$F_{\text{shear},0} = \tau dx \Big|_0 = \mu \frac{\partial u}{\partial y} \Big|_{y=0} \Delta x \quad (5)$$

$$F_{\text{shear},Y} = \tau dx \Big|_Y = \mu \frac{\partial u}{\partial y} \Big|_{y=Y} \Delta x \quad (6)$$

H.O.T. = higher order terms

Putting everything together, we obtain:

$$\begin{aligned} \int_0^Y \rho u^2 dy \Big|_{x+\Delta x} - \int_0^Y \rho u^2 dy \Big|_x + (\rho v u) \Big|_Y \Delta x \\ = - \left(\frac{\partial \rho}{\partial x} \Big|_x \Delta x + \text{H.O.T.} \right) Y - \mu \frac{\partial u}{\partial y} \Big|_0 \Delta x + \mu \frac{\partial u}{\partial y} \Big|_Y \Delta x \end{aligned}$$

We can Taylor expand our first two terms to obtain

$$\left[\frac{d}{dx} \int_0^Y \rho u^2 dy \Big|_x \Delta x + \text{H.O.T.} \right] + (\rho v u) \Big|_Y \Delta x = \Sigma F$$

But note we can divide both sides by Δx , and let $\Delta x \rightarrow 0$. All of our H.O.T. will drop. We obtain:

$$\frac{\partial}{\partial x} \int_0^Y \rho u^2 dy + (\rho v u) \Big|_Y = - \frac{\partial \rho}{\partial x} Y - \mu \frac{\partial u}{\partial y} \Big|_0 + \mu \frac{\partial u}{\partial y} \Big|_Y \quad (7)$$

The second term above is a little tricky since we don't know v . From mass conservation:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \Rightarrow \int_0^Y \frac{\partial v}{\partial y} dy = \int_0^Y - \frac{\partial u}{\partial x} dy \Rightarrow v \Big|_Y - v \Big|_0 = \int_0^Y - \frac{\partial u}{\partial x} dy$$

Now we know $v_0 = 0$ (Impermeable wall)

$$v|_y = -\int_0^Y \frac{\partial u}{\partial x} dy = -\frac{\partial}{\partial x} \int_0^Y u dy$$

Now our M_y term becomes:

$$\begin{aligned} M_y &= -\rho \frac{\partial}{\partial x} \left(\int_0^Y u dy \cdot u|_Y \right) \Rightarrow \text{Assuming } Y > \delta, u|_Y = U_\infty \\ &= -\rho \frac{\partial}{\partial x} \int_0^Y u dy \cdot U_\infty = -\rho \frac{\partial U_\infty}{\partial x} \int_0^Y u dy - U_\infty \rho \frac{\partial}{\partial x} \int_0^Y u dy \end{aligned}$$

Back substituting M_y into (7), we obtain:

$$\frac{\partial}{\partial x} \int_0^Y \rho u^2 dy - \rho \frac{\partial U_\infty}{\partial x} \int_0^Y u dy - U_\infty \rho \frac{\partial}{\partial x} \int_0^Y u dy = -\frac{\partial \rho}{\partial x} Y - \rho u \frac{\partial u}{\partial y} \Big|_0 + \rho u \frac{\partial u}{\partial y} \Big|_Y$$

Again, since $Y > \delta$, $\frac{\partial u}{\partial y} \Big|_Y = 0$ ($u = U_\infty \neq f(y)$)

$$\boxed{\frac{\partial}{\partial x} \int_0^Y u(U_\infty - u) dy = \frac{1}{\rho} \frac{\partial \rho}{\partial x} Y + \nu \frac{\partial u}{\partial y} \Big|_0 - \frac{\partial U_\infty}{\partial x} \int_0^Y u dy} \quad (8)$$

↳ Momentum Integral Boundary Layer Eqn.

Note in ME420 we automatically assumed $U_\infty = \text{constant}$ and $\frac{\partial \rho}{\partial x} = 0$, so reduces to:

$$\boxed{\frac{\partial}{\partial x} \int_0^Y u(U_\infty - u) dy = \nu \frac{\partial u}{\partial y} \Big|_0} \quad (9) \begin{aligned} &\Rightarrow \text{Flat Plate} \\ &\Rightarrow U_\infty = \text{constant}, \frac{\partial \rho}{\partial x} = 0 \end{aligned}$$

To solve this equation, we can first non-dimensionalize

$$\frac{u}{U_\infty} = \phi(\eta); \quad \eta = \frac{y}{\delta} \Rightarrow dy = \delta d\eta$$

Back substituting into (9) for a flat plate and $\frac{\partial \rho}{\partial x} = 0$

Note also, we will integrate from 0 to $Y = \delta$ since for $y > \delta$, our integral is zero (not useful to us).

$$\frac{\partial}{\partial x} \int_0^{\delta} u(u_{\infty} - u) dy = \nu \left. \frac{\partial u}{\partial y} \right|_0$$

$$u = u_{\infty} \phi$$

$$dy = \delta d\eta$$

$$\frac{\partial u}{\partial y} = \frac{u_{\infty} \partial \phi}{\delta \partial \eta}$$

$$\frac{\partial}{\partial x} \int_0^{\delta} u_{\infty} \phi (u_{\infty} - u_{\infty} \phi) \delta d\eta = \nu \frac{u_{\infty}}{\delta} \left. \frac{\partial \phi}{\partial \eta} \right|_0$$

$$\frac{\partial}{\partial x} \int_0^{\delta} u_{\infty}^2 \phi (1 - \phi) \delta d\eta = \nu \frac{u_{\infty}}{\delta} \left. \frac{\partial \phi}{\partial \eta} \right|_0$$

Since $u_{\infty} = \text{constant}$ and $\delta = f(y)$ only $f(x)$ we take it out

$$\delta \frac{\partial \delta}{\partial x} \int_0^1 \phi (1 - \phi) d\eta = \frac{\nu}{u_{\infty}} \left. \frac{\partial \phi}{\partial \eta} \right|_0$$

So our equation becomes:

$$\boxed{\delta \frac{\partial \delta}{\partial x} = \frac{\nu}{u_{\infty}} \beta} ; \quad \boxed{\beta = \frac{\phi'|_0}{\int_0^1 \phi (1 - \phi) d\eta}} \quad (10)$$

We know $\beta = f(y)$ and $\beta \neq f(x)$, so we can simply integrate

$$\int_0^x \delta d\delta = \int_0^x \frac{\nu}{u_{\infty}} \beta dx$$

$$\left. \frac{\delta^2}{2} \right|_0^x = \frac{\nu \beta x}{u_{\infty}} \Rightarrow \left(\frac{\delta}{x} \right)^2 = 2\beta \frac{\nu}{u_{\infty} x} \Rightarrow \frac{\delta}{x} = \sqrt{\frac{2\beta}{Re_x}} \quad (11)$$

We've obtained this result before with $\sqrt{2\beta} = 5.0$ (Blasius)

The way to solve momentum integral problems is to approximate the dimensionless velocity profile $\phi(\eta)$. This method is approximate, but it turns out to work very well as long as the B.C.'s are matched.

Note: It was Von-Karman (Postdoc) and Pohlhausen (student) of Prandtl who came up with this method in 1919.

Let's assume some profiles & solve for β :

$$\phi(0) = \frac{u}{u_\infty} \Big|_0 = 0 \text{ (No slip)} \quad (12)$$

$$\phi(1) = \frac{u_\infty}{u_\infty} \Big|_{\eta=1} = 1 \text{ (Free stream)} \quad (13)$$

$$\phi'(1) = \frac{\partial \phi}{\partial \eta} \Big|_{\eta=1} = 0 \text{ (no shear)} \quad (14)$$

$$\phi''(0) = \frac{\partial^2 \phi}{\partial \eta^2} \Big|_{\eta=0} = 0 \text{ (linear grad.)} \quad (15)$$

} Boundary Conditions

} $\Rightarrow$ Constant wall shear stress.

Assuming the following velocity profiles:

$$\phi = \eta \Rightarrow \text{Satisfies (12) \& (13), but not (14) or (15)}$$

$$\phi = 2\eta - \eta^2 \Rightarrow \text{Satisfies (12), (13) \& (14), but not (15)}$$

$$\phi = \sin\left(\frac{\pi}{2}\eta\right) \Rightarrow \text{Satisfies all b.c.'s}$$

$$\phi = \frac{3}{2}\eta - \frac{1}{2}\eta^3 \Rightarrow \text{Satisfies all b.c.'s}$$

Back substituting ϕ into (10) and solving for $\sqrt{2\beta'}$

$$\beta = \frac{\phi'(0)}{\int_0^1 \phi(1-\phi) d\eta}$$

ϕ	$\sqrt{2\beta}$
η	3.464
$2\eta - \eta^2$	5.477
$\sin(\pi/2 \eta)$	4.795
$\frac{3}{2}\eta - \frac{1}{2}\eta^2$	4.641

All solutions are remarkably close to $\sqrt{2\beta} = 5.0$ (Blasius exact solution!)

This shows that the integral method is particularly good at approximating the correct solution from fairly crude profile assumptions.

Let's solve for shear (τ) using $\phi = \frac{3}{2}\eta - \frac{1}{2}\eta^3$

$$\tau(x) = \mu \left. \frac{\partial u}{\partial y} \right|_0 \Rightarrow u = U_\infty \phi$$

$$dy = \delta d\eta$$

$$= \mu \left. \frac{U_\infty \partial \phi}{\delta \partial \eta} \right|_0 \Rightarrow \left. \frac{\partial \phi}{\partial \eta} \right|_{\eta=0} = \left. \frac{3}{2} - \frac{3}{2}\eta^2 \right|_0 = \frac{3}{2}$$

$$\tau(x) = \mu \frac{U_\infty}{\delta} \cdot \frac{3}{2}$$

Now if we want $C_{f,x}$

$$C_{f,x} = \frac{\tau(x)}{\frac{1}{2} \rho U_\infty^2} = \frac{\mu U_\infty 6}{2 \delta \rho U_\infty^2} = \frac{3\mu}{\delta \rho U_\infty} \left(\frac{x}{x} \right) = \frac{3\mu}{\left(\frac{\delta}{x} \right) \rho U_\infty x}$$

↗ Multiply by $\left(\frac{x}{x} \right) \Rightarrow$ trick

$$\text{But note: } Re_x = \frac{U_\infty x}{\nu} \Rightarrow C_{f,x} = \frac{3}{\frac{\delta}{x} \cdot Re_x}$$

But we already showed $\left(\frac{\delta}{x} \right) = \left(\frac{2\beta}{Re_x} \right)^{1/2} \Rightarrow$ Back substitute

$$C_{f,x} = \frac{3}{\sqrt{2\beta}} \cdot \frac{1}{\sqrt{Re_x}} \Rightarrow \text{We solved above that } \sqrt{2\beta} = 4.641$$

$$\boxed{C_{f,x} = \frac{\tau(x)}{\frac{1}{2} \rho U_\infty^2} = 0.646 Re_x^{-1/2}} \Rightarrow \text{Using similarity, we had:}$$

$$C_{f,x} = 0.664 Re_x^{-1/2}$$